

E-Commerce Sales Performance Analysis Report

(UK-Based Online Retail Dataset | Dec 2009 – Dec 2010)

1. Introduction (What & Why)

This project was carried out as part of a task-based data analytics exercise, with the goal of working with raw e-commerce sales data and transforming it into meaningful business insights.

Instead of directly creating charts, the focus was on understanding the dataset, identifying data quality issues, and determining the data relevant for accurate sales analysis. The final outcome is an interactive dashboard that enables stakeholders to quickly understand sales performance, demand patterns, and key revenue drivers.

2. Understanding the Dataset (Initial Observations)

The dataset contains online retail transaction data from a UK-based e-commerce business, covering the period from December 2009 to December 2010.

During the initial exploration, I observed:

- Multiple non-sales records such as adjustments, bad debt entries, bank charges, and manual corrections.
- Negative and zero quantities, which indicate returns, cancellations, or internal adjustments.
- Several high-value invoices with missing customer IDs.
- The year 2009 contains only December, while 2010 contains full monthly data.

These observations directly influenced how I approached data cleaning and analysis.

3. Data Cleaning & Preprocessing (How & Why)

This was the most important phase of the task.

3.1 Removal of Non-Sales Transactions

Rows with descriptions such as:

- Adjustment
- Adjust bad debt
- Amazon fees
- Bank charges
- Manual entries

were removed.

Why?

These records do not represent actual customer purchases and would distort revenue, quantity, and product performance analysis.

3.2 Handling Negative and Zero Quantity

All rows with:

- Quantity = 0
- Quantity < 0

were removed.

Reason: Negative quantities usually represent returns or corrections, not real demand. Since the task focuses on sales performance, keeping these rows would misrepresent actual customer behavior.

3.3 Date & Time Processing

The original datetime column was split into:

- Invoice Date
- Invoice Time
- Hour (derived from time)

This enabled:

- Monthly trend analysis
- Hourly sales and quantity analysis

3.4 Revenue Calculation

A new column Revenue was created:

- $\text{Revenue} = \text{Quantity} \times \text{Unit Price}$

This became the core metric for most analyses.

3.5 Customer ID Handling

Some invoices with blank customer IDs were retained.

Why?

Even without customer identification, these transactions still:

- Generated revenue
- Contributed to demand trends

Removing them would understate total sales.

4. Exploratory Data Analysis (EDA)

After cleaning, I performed EDA to understand patterns and outliers.

Key EDA Findings:

- The dataset effectively represents full-year behavior only for 2010.
- Monthly trends were analyzed across available data, with awareness of partial data in 2009.
- United Kingdom dominates in: Number of invoices, Quantity sold, Revenue generated.

5. Key Business Insights

5.1 Product Performance

- Highest revenue-generating product: Regency Cake Stand 3 Tier
- Highest quantity sold product: White Hanging Heart T-Light Holder

This shows that revenue leaders and volume leaders are not always the same, which is important for pricing and inventory strategy.

5.2 Time-Based Insights

- November recorded the highest revenue and quantity sold.
- 12 PM is the peak hour for revenue.
- PM is the peak hour for quantity sold.

This suggests strong midday purchasing behavior, useful for promotions and staffing decisions.

5.3 Invoice-Level Insight

Some invoices with a missing customer ID generated unusually high revenue, indicating bulk or wholesale purchasing behavior.

5.4 Country Performance

Only the top 5 countries were analyzed to keep insights focused.

- United Kingdom is the best-performing market by a large margin.

6. Dashboard Design (What I Chose & Why)

The dashboard was designed to be:

- Simple
- Non-technical user friendly
- Insight-focused

Instead of KPIs, Cards were used to clearly show:

- Total Sales Revenue
- Total Quantity Sold
- Total Invoices

Visuals include:

- Monthly Sales & Quantity Distribution (include both the year)
- Hourly Sales vs Quantity Distribution
- Top Performing Products by Sales & Quantity
- Country-wise Performance (Top 5)

Total Sales Revenue

£10.01M

Sum of Total Sales

Total Quantity Sold

5.83M

Sum of Quantity sold

Total Invoices

20.67K

Count of Invoice

E-Commerce Sales Performance Dashboard

(UK Retail | Dec 2009 - Dec 2010)

Year

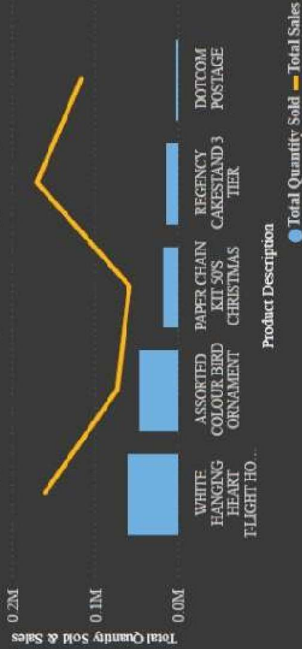
2009

2010

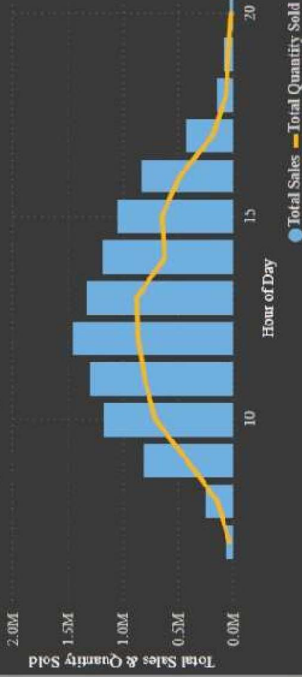
Month

January	March	May	July	September	November
February	April	June	August	October	December

Top Performing Products by Sales & Quantity



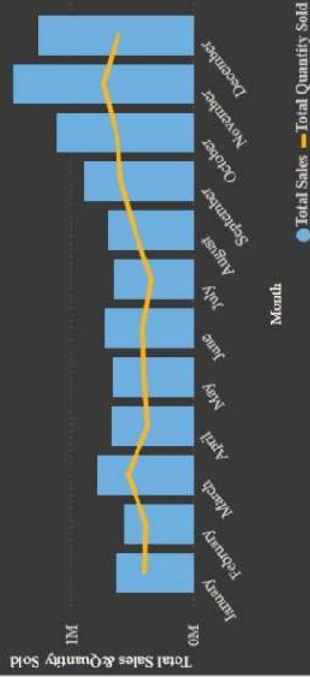
Hourly Sales & Quantity Distribution



Country-wise Sales Performance



Monthly Sales & Quantity Distribution



7. Conclusion

This project demonstrates the end-to-end analytics workflow:

- Understanding messy real-world data
- Cleaning with business logic
- Performing meaningful EDA
- Translating insights into a dashboard

The task strengthened my ability to think like a business-focused data analyst, not just a tool user.